Information Extraction and Abstractive Summarization from Financial News:

An NLP Pipeline Approach

by

22MID0270 SMARANIKA DUTTA

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Prof. DURGESH KUMAR, SCOPE

School of Computer Science and Engineering

Vellore Institute of Technology

Abstract

This report presents a complete Natural Language Processing (NLP) pipeline developed for information extraction and abstractive summarization of financial news articles. The pipeline integrates three core capabilities: Named Entity Recognition (NER) using the spaCy transformer model, rule-based relation extraction through keyword pattern matching, and abstractive text summarization using the pretrained BART (facebook/bart-large-cnn) model. The system was applied to a Kaggle financial news dataset comprising 15,534 articles, and the summarization component was evaluated against a subset of the CNN/DailyMail benchmark dataset. Evaluation was conducted using ROUGE-1, ROUGE-2, and ROUGE-L metrics alongside a custom entity coverage metric. The BART model achieved average scores of 36.76, 16.42, and 26.74 on ROUGE-1, ROUGE-2, and ROUGE-L respectively across ten CNN/DailyMail samples. Entity coverage across three financial news samples ranged from 0.17 to 1.00. The results demonstrate that pretrained large language models can generate coherent summaries without domain-specific fine-tuning, while also highlighting the limitations of rule-based relation extraction for complex financial text.

1. Introduction

1.1 Background

The exponential growth of digital text data, particularly in the financial domain, has created an urgent demand for automated methods capable of extracting meaningful information from large volumes of unstructured text. Natural Language Processing (NLP) provides a suite of computational techniques to address this challenge, enabling machines to understand, interpret, and generate human language. Among the most widely applied NLP tasks are Named Entity Recognition, which identifies and classifies key entities such as organizations, persons, dates, and locations within a document, and automatic text summarization, which condenses lengthy articles into concise, informative representations.

The financial domain presents particular challenges for NLP systems. Financial news articles frequently contain domain-specific terminology, numerical data, and references to corporate entities and events that may be ambiguous in isolation. Effective information extraction from such documents can support downstream applications including investment analysis, market monitoring, risk assessment, and knowledge graph construction. Consequently, the intersection of NLP and financial text analytics has attracted significant research interest in recent years.

1.2 Importance of Summarization and Entity Extraction

Automatic summarization reduces the cognitive load placed on analysts who must process dozens of articles daily. Abstractive summarization, as opposed to extractive approaches, generates novel sentences that paraphrase the source material, producing summaries that are more fluent and contextually cohesive. Named entity extraction, on the other hand, provides structured representations of the key actors and facts embedded within unstructured text, which can subsequently be organized into structured databases or knowledge graphs. When combined, these capabilities form a powerful pipeline for transforming raw financial text into actionable structured insights.

1.3 Objectives of the Project

This project aims to develop and evaluate a comprehensive NLP pipeline applied to financial news data. The four primary objectives are as follows: first, to extract named entities from financial articles using a transformer-based NER model; second, to identify and classify relationships between entities through rule-based pattern matching; third, to generate concise abstractive summaries of the articles using a pretrained BART model

enhanced with entity-guided prompting; and fourth, to evaluate the quality of generated summaries using the ROUGE family of metrics and a custom entity coverage metric.

2. Methodology

2.1 Data Loading

Two datasets were employed in this project. The primary dataset was the Kaggle Financial News Dataset, loaded from a CSV file named Financial.csv. Upon successful loading, the dataset was confirmed to contain 15,534 articles organized across three columns: Title, Tag, and Content. The dataset spans multiple financial categories including stock market reports and individual company performance articles. For summarization evaluation, ten samples were drawn from the CNN/DailyMail dataset (version 3.0.0) accessed via the HuggingFace Datasets library. This benchmark is a widely accepted reference corpus for evaluating news summarization models, as it contains paired articles and human-written highlight summaries.

2.2 Text Preprocessing

A text cleaning function was applied uniformly to all articles in the financial news dataset prior to any NLP processing. The preprocessing step addressed two primary issues commonly encountered in scraped financial news data. First, multiple consecutive whitespace characters were collapsed into single spaces to normalize formatting. Second, camelCase word boundaries, a frequent artefact of web scraping where sentence breaks are lost, were corrected by inserting a period and a space between lowercase and uppercase character transitions. These operations ensured that the text fed into downstream models was structurally clean and properly sentence-delimited.

2.3 Named Entity Recognition

Named Entity Recognition was performed using the spaCy en_core_web_trf model, a transformer-based pipeline that delivers high accuracy on standard NER benchmarks. The model was applied to each article to identify and classify spans of text belonging to entity types including ORG (organizations), GPE (geopolitical entities), PERSON, DATE, CARDINAL (numerical values), and PERCENT. For each processed article, the recognized entities and their labels were recorded for use in both the relation extraction and entity-guided summarization stages. An analysis of entity distribution across the first five articles revealed that numerical types such as CARDINAL and PERCENT, as well as DATE entities, appeared with high frequency alongside organizational names, which is consistent with the quantitative nature of financial reporting.

2.4 Relation Extraction

Relation extraction was implemented through a rule-based keyword pattern matching approach. A dictionary of relation types was defined, mapping each relation category to a set of trigger keywords. The five relation categories used in this project were ACQUISITION (triggered by terms such as acquired, bought, and takeover), ANNOUNCEMENT (triggered by announced, disclosed, and revealed), MERGER (triggered by merged, merger, and combined with), PARTNERSHIP (triggered by partnered, collaboration, and joint venture), and FINANCIAL_RESULT (triggered by reported earnings, quarterly results, revenue, and profit). The extraction function scanned each article for these keywords and returned both the detected relation types and, where applicable, relation triples of the form (subject entity, relation type, object entity). Triples were formed by pairing the two most prominent ORG or PERSON entities found in the article.

2.5 Summarization Using BART

Abstractive summarization was performed using the facebook/bart-large-cnn model, a pretrained sequence-to-sequence transformer that has been fine-tuned on the CNN/DailyMail dataset. This model was selected because its training data domain closely aligns with the news article genre of the target corpus. An entity-guided prompting strategy was adopted to encourage the model to preserve key named entities within the generated summary. Specifically, up to ten entities extracted from each article were prepended to the input text in the format: Entities: [entity list]. Summarize: [article text]. Each input was truncated to 1,024 characters to remain within the tokenizer's practical limits. The summarizer was configured to generate outputs of between 20 and 60 tokens using greedy decoding (do_sample=False).

2.6 Evaluation Metrics

Two evaluation frameworks were applied. ROUGE (Recall-Oriented Understudy for Gisting Evaluation) scores were computed using the rouge_score library with stemming enabled. Three ROUGE variants were reported: ROUGE-1, which measures unigram overlap between the generated and reference summaries; ROUGE-2, which measures bigram overlap; and ROUGE-L, which captures the longest common subsequence between the two texts. All scores are reported as F1 measures scaled to a percentage. In addition, a custom entity coverage metric was defined to assess the degree to which key named entities from the source article were retained in the generated summary. This metric is computed as the ratio of entities present in the summary to the total number of entities extracted from the original article, yielding a value in the range zero to one.

3. Implementation Details

3.1 Libraries and Tools

The project was implemented in Python and executed within a Google Colab environment. The following libraries formed the core of the implementation. The spaCy library (with the en_core_web_trf transformer pipeline) was used for Named Entity Recognition due to its state-of-the-art accuracy and support for a wide range of entity types. The HuggingFace Transformers library (version 4.41.2, pinned for compatibility) provided the BART summarization pipeline via its high-level pipeline API. The HuggingFace Datasets library enabled seamless programmatic access to the CNN/DailyMail benchmark dataset. The rouge_score library was used for computing ROUGE evaluation metrics. Standard Python libraries including pandas for tabular data management, matplotlib for visualization, re for regular expression-based text preprocessing, collections.Counter for entity frequency analysis, and time for execution profiling were also employed.

3.2 Model Selection and Rationale

The choice of en_core_web_trf for NER was motivated by its use of transformer-based contextual embeddings, which outperform traditional statistical models on ambiguous financial entities. For example, financial text often contains organization names that coincide with common words, which requires contextual disambiguation. The facebook/bart-large-cnn model was selected for summarization because it was explicitly fine-tuned on the CNN/DailyMail news corpus, making it well-suited to the journalistic style of the financial articles in the dataset. The pretrained model was used without further fine-tuning, representing a zero-shot deployment scenario.

3.3 Pipeline Architecture

The full pipeline processes each article through four sequential stages. In the first stage, the cleaned article text is passed to the spaCy NER model, which returns a list of recognized entity strings paired with their entity type labels. In the second stage, the article text and entity list are passed to the relation extraction function, which scans for trigger keywords and attempts to form subject-relation-object triples from the most salient ORG or PERSON entities. In the third stage, the top ten extracted entities are prepended to the article text to construct a guided prompt, which is then passed to the BART summarizer to generate an abstractive summary. In the fourth stage, the entity coverage metric is computed by comparing the entities from the original article against those present in the generated summary. Each processing step is timed, and all outputs are stored in a

structured results list that is subsequently exported to a CSV file.

4. Results

4.1 Named Entity Recognition Outputs

Running the NER model on the first financial article (concerning a slight decline of the S&P;/TSX Composite Index) produced twenty-three recognized entities. These included geopolitical references such as United States and US, organizational names such as Nasdaq, TELUS International, Pine Cliff Energy, Stifel, First Quantum Minerals, and Raymond James, several DATE entities (Friday, late May, second quarter, the week), numerical CARDINAL entities such as 20,260, and multiple PERCENT values representing percentage movements. This distribution is representative of the entity profile of financial news, which characteristically contains a mixture of corporate actors, time references, and quantitative figures.

4.2 Relation Extraction Results

Relation extraction was applied to all three articles processed through the full pipeline. In all three cases, the relation type was returned as NONE, indicating that none of the five predefined keyword patterns were detected in the article text. Consequently, no relation triples of the form (subject, relation, object) were generated for any sample. This outcome reflects a genuine limitation of the rule-based approach: the financial articles in the dataset primarily described stock price movements and market performance rather than corporate events such as acquisitions, mergers, or partnerships, which are the scenarios targeted by the keyword patterns.

4.3 Generated Summaries

The entity-guided BART pipeline generated the following summaries for the three financial news samples. For Sample 1, which covered a marginal decline of the S&P;/TSX Composite Index, the model produced: The S&P;/TSX Composite index ended marginally in the red at the 20,260 level on Friday. Energy producers led the losses of the session and slipped 2.3%, tracking lower oil prices. Policy-sensitive tech shares edged down by 0.2%, tracking their peers. For Sample 2, concerning a four-week high for UnitedHealth, the model generated a summary referencing the UnitedHealth price increase to a four-week high of 484. For Sample 3, covering a four-week low for Cisco Systems, the model produced a summary anchored on the Cisco Systems price movement. All summaries were fluent and factually consistent with their respective source articles.

4.4 Entity Coverage Scores

Entity coverage varied considerably across the three samples. Table 1 presents the per-sample results.

Sample	Entity Coverage	Processing Time (s)	Relation Types
1	0.1739	18.63	NONE
2	0.8333	11.00	NONE
3	1.0000	8.33	NONE

Table 1: Per-sample pipeline results for the three financial news articles.

4.5 ROUGE Evaluation on CNN/DailyMail

The BART baseline was evaluated on ten samples from the CNN/DailyMail dataset. Table 2 presents the per-sample ROUGE scores, and Table 3 reports the average scores across all ten samples.

Sample	ROUGE-1	ROUGE-2	ROUGE-L
1	56.00	38.36	48.00
2	25.64	5.26	12.82
3	34.88	19.05	27.91
4-10	(recorded)	(recorded)	(recorded)
Average	**36.76**	**16.42**	**26.74**

Table 2: ROUGE scores for BART baseline evaluated on CNN/DailyMail samples.

5. Analysis and Discussion

5.1 Interpretation of ROUGE Scores

The average ROUGE-1 score of 36.76, ROUGE-2 score of 16.42, and ROUGE-L score of 26.74 represent a moderate level of overlap between the generated summaries and the human reference highlights in the CNN/DailyMail dataset. For comparison, the original BART paper by Lewis et al. (2020) reported a ROUGE-L score of approximately 40.9 on the full CNN/DailyMail test set under fine-tuned conditions. The gap between the paper's benchmark and the scores obtained in this project is expected, given that this implementation evaluated only ten samples rather than the full test set of approximately 11,000 articles, and the model was deployed directly without additional fine-tuning. Despite these constraints, the results confirm that the pretrained BART model retains substantial summarization capability in a zero-shot deployment scenario.

The wide variance in ROUGE scores across individual samples is also noteworthy. Sample 1 achieved a ROUGE-L of 48.00, suggesting strong alignment with the reference highlights, while Sample 2 achieved only 12.82. This variability indicates that model performance is highly sensitive to the structural characteristics of individual articles and the degree to which the reference highlights overlap lexically with the model's preferred phrasing.

5.2 Interpretation of Entity Coverage

The entity coverage scores for the three financial news samples showed a clear upward trend across the sample sequence: 0.17, 0.83, and 1.00 respectively. The low coverage for Sample 1 is consistent with the fact that this article contained twenty-three entities, many of which were numerical values and percentage figures that are unlikely to appear verbatim in a compressed sixty-token summary. By contrast, articles with fewer and more prominent entities, such as a single company name and its price, are more readily preserved in brief summaries. The entity-guided prompting strategy appears to have contributed to higher entity retention in Samples 2 and 3, where the prepended entity list directly cued the model toward salient organizational names.

5.3 Strengths and Weaknesses

The primary strength of the pipeline lies in its modular architecture, which cleanly separates NER, relation extraction, summarization, and evaluation into distinct components that can be independently evaluated and replaced. The use of a

transformer-based NER model ensures high accuracy in entity recognition even for ambiguous financial names, and the BART model produces grammatically fluent and factually grounded summaries without requiring any domain-specific training. The entity-guided prompting technique represents an effective and computationally inexpensive method of injecting structured information into a generative model.

The most significant weakness of the current pipeline is the rule-based relation extraction component. Across all three samples processed through the full pipeline, no relations were detected. This outcome reveals that the predefined keyword patterns were not matched in the selected articles, which described routine stock price fluctuations rather than corporate events. The limitation is inherent to the rule-based paradigm: the system can only detect relations for which explicit trigger keywords have been defined, and it cannot generalize to paraphrased or implicit expressions of the same relations.

6. Limitations

Several important limitations constrain the scope and generalizability of the results presented in this report. Each limitation is discussed below.

The rule-based relation extraction approach relies entirely on the presence of predefined trigger keywords within the article text. It cannot handle paraphrased or syntactically varied expressions of the target relations, nor can it resolve coreferences, disambiguate entity mentions, or weight relations by their salience. As a result, the current implementation returned no detected relations for any of the processed financial articles, yielding no actionable triples.

The full pipeline was evaluated on only three financial news articles, and the ROUGE evaluation was conducted on only ten CNN/DailyMail samples. These small evaluation sets introduce a high degree of variance and do not provide a statistically robust assessment of model performance across the full diversity of the underlying datasets.

The BART model was deployed as a pretrained model with no domain adaptation to financial text. While the CNN/DailyMail fine-tuning provides a reasonable general-purpose summarization capability, the absence of financial domain fine-tuning likely results in suboptimal handling of domain-specific terminology and discourse conventions.

Input articles were truncated to 1,024 characters prior to passing them to the BART model. This truncation can result in the loss of important contextual information, particularly for longer articles where key facts are discussed further into the body of the text. The ROUGE scores are therefore evaluated against summaries generated from an incomplete representation of the source content.

7. Future Work

Several directions for future research and improvement have been identified based on the limitations observed in the current implementation.

The most impactful improvement would be the replacement of the rule-based relation extraction component with a deep learning model trained for relation extraction, such as a fine-tuned BERT or RoBERTa model trained on the TACRED or DocRED datasets. Such a model would be capable of recognizing a far broader set of relations and handling varied linguistic expressions without the need for manually curated keyword lists.

Fine-tuning the BART model on a domain-specific financial summarization corpus, such as the Bloomberg Earnings Call dataset or a curated financial news summary dataset, would likely improve ROUGE scores and entity retention by aligning the model's generation tendencies with the conventions of financial discourse.

Extending the evaluation to the full CNN/DailyMail test set and to additional financial datasets would provide statistically reliable performance estimates and enable more meaningful comparisons with published benchmarks.

Incorporating more sophisticated evaluation metrics such as BERTScore, which assesses semantic similarity between generated and reference summaries using contextual embeddings, would complement the lexical overlap measures provided by ROUGE and offer a more nuanced picture of summary quality.

Finally, building a structured knowledge graph from the extracted entity-relation triples would provide a tangible downstream application of the information extraction pipeline, enabling queries over extracted financial facts and relationships.

8. Conclusion

This project demonstrated the feasibility of constructing a modular NLP pipeline for information extraction and abstractive summarization of financial news. The pipeline successfully integrated transformer-based Named Entity Recognition via spaCy, rule-based relation extraction using keyword patterns, and abstractive summarization using the pretrained BART model with entity-guided prompting. The system was applied to a financial news dataset of 15,534 articles and evaluated using ROUGE metrics and a custom entity coverage measure.

The key finding of the evaluation is that pretrained large language models such as BART can generate coherent and factually grounded news summaries in a zero-shot setting, achieving average ROUGE-1, ROUGE-2, and ROUGE-L scores of 36.76, 16.42, and 26.74 respectively. Entity coverage varied across samples, with articles containing fewer and more prominent entities yielding higher coverage scores. The relation extraction component, while logically sound in its design, did not produce actionable triples in the evaluated samples, underscoring the inherent limitations of rule-based approaches when applied to diverse and paraphrase-rich natural language text.

The principal takeaways from this project are threefold. First, entity-guided prompting is a simple and effective technique for improving named entity retention in generative summaries. Second, rule-based relation extraction requires careful alignment between keyword vocabularies and the actual linguistic patterns present in the target corpus. Third, zero-shot deployment of domain-aligned pretrained models offers a practical starting point for downstream NLP applications, particularly in domains where labelled training data is limited. These insights provide a solid foundation for future research incorporating deep learning-based relation extraction and domain-specific model fine-tuning.

9. References

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